

After Study Report

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DOCUMENT HISTORY

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1.0	2026-01-24	First version of the after study report	AF, AG, VG, MM	DN

1 INTRODUCTION

This document presents an after study report for the project. The purpose of this report is to reflect on the work carried out during the project, identify the main challenges encountered, and summarize the key lessons learned. The reflections focus on project planning, time management, and technical understanding gained throughout the implementation and testing phases.

2 TIME MANAGEMENT

One of the main challenges faced during the project was inadequate time management. The project timeline was not planned in sufficient detail at the beginning, which led to increased workload and time pressure in later stages. Certain tasks took longer than initially expected, and more time should have been allocated for planning, testing, and iteration.

With better initial planning and clearer milestones, the workload could have been distributed more evenly throughout the project duration. This would have reduced stress and allowed more time for refinement and validation of the implemented solutions.

3 PROBLEMS FACED

The primary problem encountered during the project was the lack of proper time management. Although the scope and objectives of the project were understood, the effort required for different tasks was underestimated. As a result, some activities were completed under time constraints, limiting the opportunity for deeper analysis and optimization.

Another challenge was balancing project work with other academic commitments. This further emphasized the importance of early planning and continuous progress throughout the project timeline.

4 LESSONS LEARNED

Several important lessons were learned during the course of the project:

- The importance of creating a detailed and realistic time plan at the beginning of the project.
- A better understanding of signal processing concepts, particularly how signals behave in the frequency domain when applying the Fast Fourier Transform (FFT).
- Improved knowledge of onset and offset identification in signals, and how these features can be detected and interpreted in practical applications.

These lessons contributed significantly to both technical growth and improved project management awareness.

5 ALTERNATIVE APPROACHES

If the project were to be carried out again, more emphasis would be placed on early planning and incremental development. Starting implementation earlier and validating intermediate results could have reduced last-minute pressure.

An important alternative approach would have been to conduct a more extensive literature review at an earlier stage of the project. Reading and analyzing research papers related to machine learning implementations for signal processing could have provided valuable insights and guidance. This area proved challenging during the project, and having stronger theoretical and practical references from existing work might have helped in choosing more effective methods and avoiding unnecessary difficulties.

From a technical perspective, additional experimentation with signal preprocessing techniques before applying FFT and onset/offset detection could further improve robustness and accuracy.

6 REQUIREMENT FULFILLMENT

Despite the challenges faced, the project objectives were met. The core functionalities were implemented and tested, and the system behaved as expected according to the defined requirements. The project successfully demonstrated the intended signal processing pipeline and analysis methods.

7 FINAL REMARKS

Overall, the project provided valuable hands-on experience in both technical signal processing and collaborative project work. The challenges faced highlighted the importance of planning and continuous progress, while the technical tasks deepened the understanding of FFT-based analysis and signal event detection. The experience gained during this project will be beneficial for future academic and industrial projects.